Financial Time Series
46-929
Solutions #6

1. Download from canvas the file fredgraph.csv. This file contains data from the St. Louis Federal Reserve giving the U.S. Treasury monthly average constant maturity rates for 1 year, 5 years and 10 years dating back to 1953. Plot these variables, perform a Dickey Fuller test to check for stationarity. If needed, difference the data to transform it to $\mathcal{I}(1)$. Perform cointegration tests to check if any pairs of these data are cointegrated, following what was done in class or equivalently follow the example give in

   https://bashtage.github.io/arch/doc/unitroot/unitroot_cointegration_examples.html

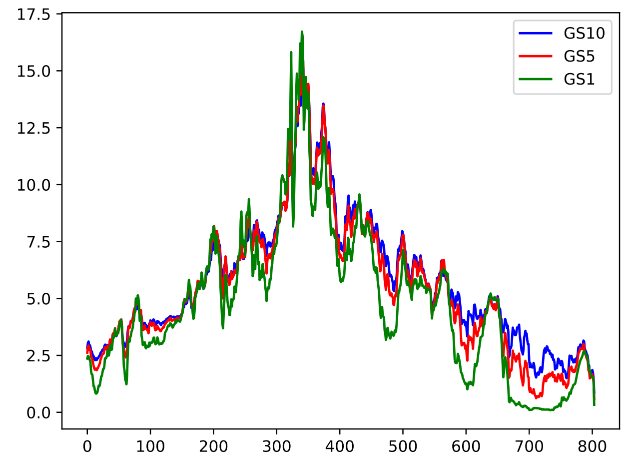
   Solution

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from arch.unitroot import engle_granger
from arch.unitroot.cointegration import phillips_ouliaris
from statsmodels.tsa.vector_ar.vecm import coint_johansen

import numpy as np
import statsmodels as smm
import statsmodels.api as sm
import statsmodels.formula.api as smf
import statsmodels.tsa as smt
import scipy.stats as scs
import matplotlib.pyplot as plt

data = pd.read_csv('fredgraph.csv')
data.head()
```



smm.tsa.stattools.adfuller(data['GS10'])
(-0.867, 0.80, 21, 782, '1%': -3.44, '5%': -2.87, '10%': -2.59, 38.87)
smm.tsa.stattools.adfuller(data['GS5'])
(-1.06, 0.73, 21, 782, '1%': -3.44, '5%': -2.87, '10%': -2.57, 235.69)
smm.tsa.stattools.adfuller(data['GS1'])
(-1.77, 0.39, 20, 783, '1%': -3.44, '5%': -2.87, '10%': -2.57, 556.53)

This shows that the H_0 in the Dickey Fuller test is NOT rejected, hence we cannot conclude that any of the three series are stationary. Hence we difference each series one time.

diff10 = np.diff(data['GS10'])
diff5 = np.diff(data['GS5'])
diff1 = np.diff(data['GS1'])
smm.tsa.stattools.adfuller(diff10)
(-6.89, 1.37e-09)
smm.tsa.stattools.adfuller(diff5)
(-6.81, 2.08e-09) smm.tsa.stattools.adfuller(diff1)
(-6.78, 2.47e-09)

In all three cases, the null hypothesis of a unit root is overwhelmingly rejected. So each of the differenced series is stationary, and the series themselves are all $\mathcal{I}(1)$.

eg_test = engle_granger(data['GS10'], data['GS5'], trend="n")
eg_test
Engle-Granger Cointegration Test
Statistic: -3.088906009822177
P-value: 0.021326184167723132
Null: No Cointegration, Alternative: Cointegration

eg_test = engle_granger(data['GS10'], data['GS1'], trend="n")
eg_test
Engle-Granger Cointegration Test
Statistic: -3.4015501707752556

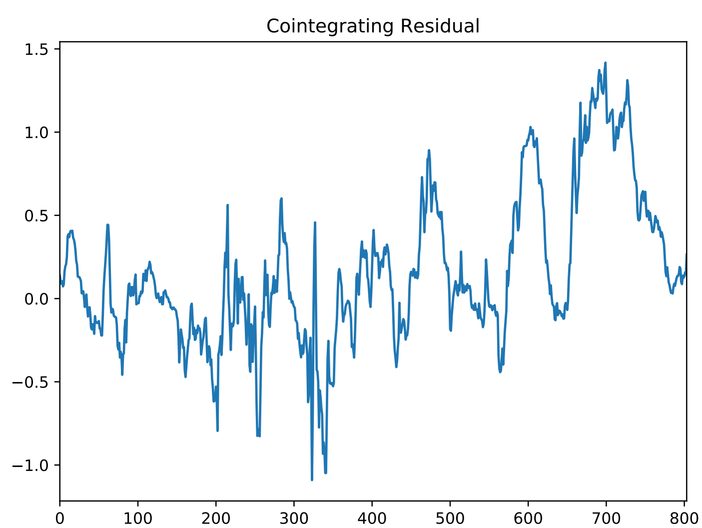
P-value: 0.008451634515862455
Null: No Cointegration, Alternative: Cointegration

eg_test = engle_granger(data[’GS5’], data[’GS1’], trend=”n”)
eg_test
Engle-Granger Cointegration Test
Statistic: -3.8647927995305857
P-value: 0.0017883424447213106
Null: No Cointegration, Alternative: Cointegration

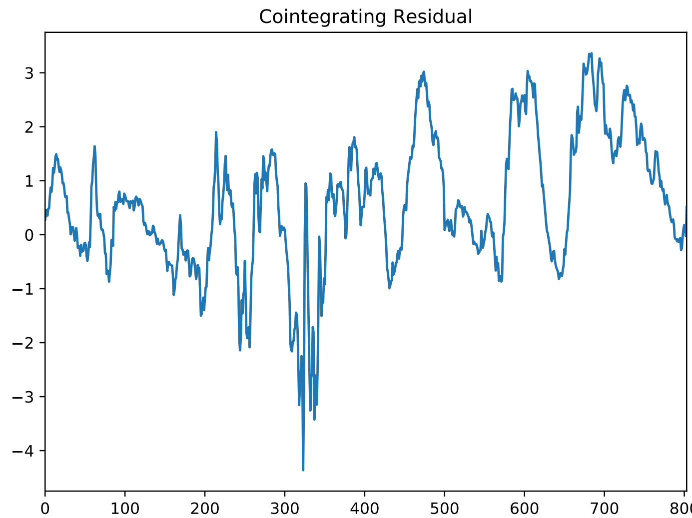
fig = eg_test.plot()
eg_test.cointegrating_vector
GS10 1.000000
GS5 -1.027517

fig = eg_test.plot()
plt.show()
eg_test.cointegrating_vector
GS10 1.000000
GS1 -1.081807

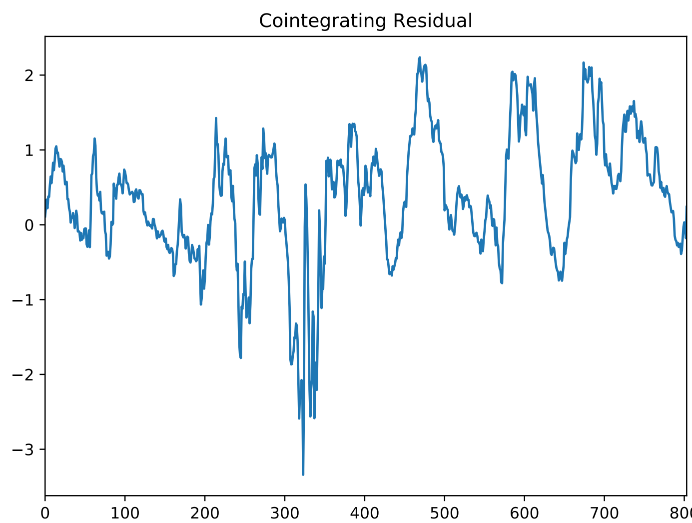
fig = eg_test.plot()
plt.show()
eg_test.cointegrating_vector
GS5 1.000000
GS1 -1.062635



Cointegration Residuals: GS10 and GS5



Cointegration Residuals: GS10 and GS1



Cointegration Residuals: GS5 and GS1

These Engle-Granger test show that all pairs of series are cointegrated, it is nearly the case that differencing any of the three pairs gives a stationary process as the cointegrating vectors are all approximately (1,-1). The GS10 and GS5 pair has a p-value of .02, hence we can reject the lack of cointegration. The other two pairs have smaller p-values.

po_zt_test = phillips_ouliaris(data['GS10'], data['GS5'], trend="c", test_type="Zt")
po_zt_test.summary()
¡class 'statsmodels.iolib.summary.Summary'¿

Phillips-Ouliaris zt Cointegration Test
=====================================
Test Statistic -3.398
P-value 0.043

Kernel Bartlett
Bandwidth 12.829
————————————-

Trend: Constant Critical Values: -3.05 (10Null Hypothesis: No Cointegration Alternative Hypothesis: Cointegration Distribution Order: 3

po_zt_test = phillips_ouliaris(data['GS10'], data['GS1'], trend="c", test_type="Zt")
po
zt_test.summary()
¡class 'statsmodels.iolib.summary.Summary'¿

Phillips-Ouliaris zt Cointegration Test
=======================================
Test Statistic -3.692
P-value 0.019
Kernel Bartlett
Bandwidth 11.545
————————————-

Trend: Constant
Critical Values: -3.05 (10%), -3.35 (5%), -3.92 (1%)
Null Hypothesis: No Cointegration
Alternative Hypothesis: Cointegration
Distribution Order: 3

po_zt_test = phillips_ouliaris(data['GS5'], data['GS1'], trend="c", test_type="Zt")
po_zt_test.summary()
¡class 'statsmodels.iolib.summary.Summary'¿

Phillips-Ouliaris zt Cointegration Test
=======================================
Test Statistic -4.244
P-value 0.003
Kernel Bartlett
Bandwidth 16.167
————————————-

Trend: Constant
Critical Values: -3.05 (10%), -3.35 (5%), -3.92 (1%)
Null Hypothesis: No Cointegration
Alternative Hypothesis: Cointegration
Distribution Order: 3

The Phillips-Ouliaris test also show cointegration of all the pairs, although the p-value for the GS10 and GS5 pair has a p-value of .04 and so is barely significant. The others are more clearly significant, thus the Phillips-Ouliaris test agrees with the Engle Granger test. The Johansen test was also run for both the trace test and the maximum eigenvalue test.

```
johansen_rst = coint_johansen(data, det_order = 0, k_ar_diff = 0)
johansen_rst.lr1
trace statistic: array([53.44485201, 18.08751136, 3.65095396])
johansen_rst.lr2
maximum eigenvalue statistic: array([35.35734065, 14.43655739, 3.65095396])

johansen_rst.cvt
array(((27.0669, 29.7961, 35.4628),
(13.4294, 15.4943, 19.9349),
( 2.7055, 3.8415, 6.6349)))
johansen_rst.cvm
array(((18.8928, 21.1314, 25.865 ),
(12.2971, 14.2639, 18.52 ),
( 2.7055, 3.8415, 6.6349)))
```

According to the trace statistic there is one cointegrating vector at the 1% level and two at the 5% level. The same is true using the maximum eigenvalue statistic.

2. Consider a univariate AR(p) process

$$X_t - \mu = \phi_1(X_{t-1} - \mu) + \cdots \phi_p(X_{t-p} - \mu) + \epsilon_t.$$

Write this is state space form, identifying all of the components of that form taking $\mathbf{X}_t = (X_t - \mu, \cdots, X_{t+1-p} - \mu)$ (p-dimensional) and X_t (1-dimensional) as the variable in the observation equation.

Solution
We can put the AR(1) model into state space form for the processes

$$\mathbf{X}_t = \mathbf{F}_t\mathbf{X}_{t-1} + \mathbf{V}_t,$$

$$\mathbf{Y}_t = \mathbf{G}_t\mathbf{X}_t + \mathbf{W}_t,$$

with $Y_t = (1, \cdots, 0)\mathbf{X}_t$, $\mathbf{W}_t = \mathbf{0}$, $\mathbf{V}_t = (\epsilon_t + \mu(1 - \sum_{i=1}^{p} \phi_i), 0, \cdots, 0)^T$ $\mathbf{X}_t = (X_t, \cdots, X_{t+1-p})^T$ and

$$F = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 & \cdots & \phi_p \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & & 0 \\ 0 & & & 1 & 0 \end{bmatrix}.$$

3. Consider the local trend model

$$M_{t+1} = M_t + V_t,$$

$$Y_t = M_t + W_t,$$

where $\{V_t\}$ is $\mathrm{WN}(0, \sigma_V^2)$ and $\{W_t\}$ is $\mathrm{WN}(0, \sigma_W^2)$

a) Show that Y_t can be written as an ARIMA(0,1,1) model

$$Y_t = Y_{t-1} + Z_t + \theta Z_{t-1},$$

for an appropriately chosen noise sequence Z_t, where $\{Z_t\}$ is $\mathrm{WN}(0, \sigma^2)$.

b) Determine possible solutions for θ and σ in terms of σ_V^2 and σ_W^2.

c) Let $\sigma_W^2 = 8$ and $\sigma_V^2 = 20$, simulate a series of Y_t according to this local trend model for the possible solutions found in part b) with these specific values for σ_W^2 and σ_V^2.

d) Perform a time series analysis on the simulated Y_t series. What do you conclude?

Solution

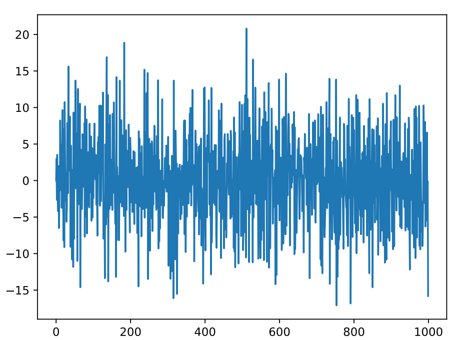
Note, Problem 2 is about putting a standard time series model into state space form. This problem is taking a model in state space form and putting it into traditional form. There are two possible solutions.

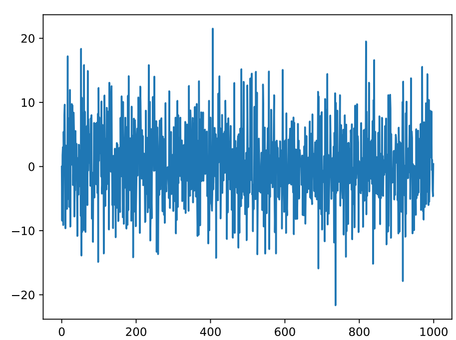
a) To show that Y_t is an ARIMA(0,1,1) series, look at the autocorrelation of the first difference series. We first determine the autocovariance function of the $\nabla Y_t = Y_t - Y_{t-1}$ process. $Y_t - Y_{t-1} = M_t - M_{t-1} + W_t - W_{t-1} = V_{t-1} + W_t - W_{t-1}$. $\gamma(0) = Var(V_{t-1} + W_t - W_{t-1}) = \sigma_V^2 + 2\sigma_W^2$. Similarly, $\gamma(1) = Cov(V_{t-1} + W_t - W_{t-1}, V_{t-2} + W_{t-1} - W_{t-2}) = -\sigma_W^2$. Note that $\gamma(h) = 0$ for $h \geq 2$. Now, for a MA(1) process where $Y_t - Y_{t-1} = Z_t + \theta Z_{t-1}$, $\gamma(0) = Var(Z_t + \theta Z_{t-1}) = \sigma^2(1 + \theta^2)$. $\gamma(1) = Cov(Z_t + \theta Z_{t-1}, Z_{t-1} + \theta Z_{t-2}) = \theta\sigma^2$. Furthermore, $\gamma(h) = 0$ for $h \geq 2$.

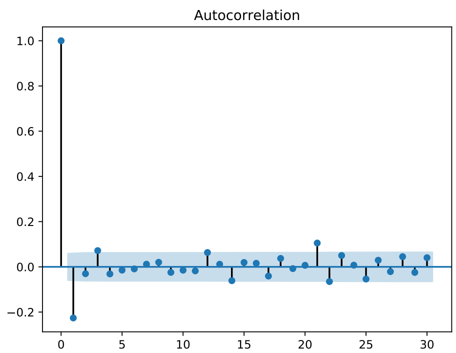
b) We have two simultaneous equations, $\theta\sigma^2 = -\sigma_W^2$ and $(\theta^2 + 1)\sigma^2 = \sigma_V^2 + \sigma_W^2$. Substituting $\sigma^2 = -\sigma_W^2/\theta$ we find that θ must satisfy a quadratic equation $(1 + \theta^2)/\theta = -((\sigma_V^2 + 2\sigma_W^2)/\sigma_W^2$. This is a simple quadratic equation that can be explicitly solved for (θ, σ^2) in terms of (σ_V^2, σ_W^2). There will be two real solutions.

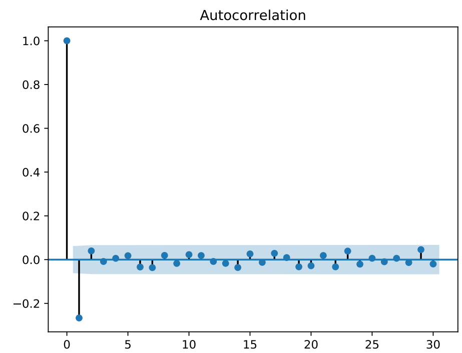
c) If $\sigma_W^2 = 8$ and $\sigma_V^2 = 20$, there are two solutions $(\sigma^2 = 1.8754, \theta = -4.2656)$ and $(\sigma^2 = 34.1245, \theta = -.2344)$.

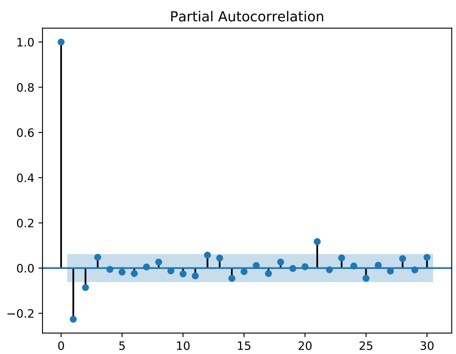
d) Although the two parameter sets are quite different, the time series displays (time series plot, ACF and PACF) are essentially indistinguishable looking just at the second order properties.

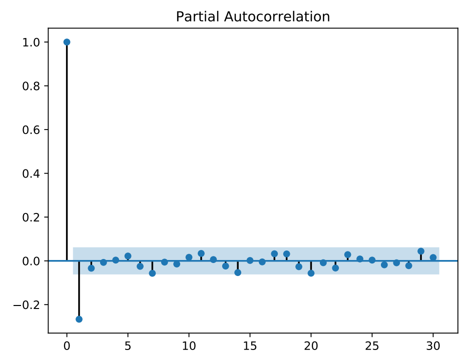





Autocorrelation


Autocorrelation


Partial Autocorrelation


Partial Autocorrelation